

Sree Bhargavi Balija

Computer Science PhD Researcher

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SUMMARY

- Machine Learning Researcher & Engineer with 3+ years of experience in scalable ML systems, federated learning, and generative AI, backed by 8 published papers (including work collaborated with **MIT Media Lab**) and ongoing research with **Stanford University's** Prof. Ludwig on **Terminal bench** Project.
 - Built RAG systems and fine-tuned GPT-4/LLaMA-2 using LangChain, Hugging Face, and Pinecone/Weaviate. Automated workflows with multi-agent frameworks, cutting manual effort by 40%+ via document AI and autonomous agents.
 - Spearheaded **quantitative modeling** (XGBoost, LightGBM, Prophet) and privacy-preserving AI via federated learning (TensorFlow **Federated**, Flower). Deployed event-driven microservices (**Kafka**, **Pulsar**) for real-time fraud detection and risk analytics, improving prediction accuracy by 25% while ensuring SOC-2/GDPR compliance.
 - Designed end-to-end MLOps pipelines with CI/CD (**GitHub Actions**, **ArgoCD**), model monitoring (Evidently, Prometheus), and feature stores (**Feast**). Enhanced latency-critical APIs (Python/Java) using FastAPI, **Spring Boot**, and gRPC, achieving 99.9% uptime for systems serving 100K+ users with **Elasticsearch/Cassandra** for low-latency analytics.
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PROFESSIONAL EXPERIENCE

Software Engineering Intern Fujitsu Research of America

USA

Aug 2025 – Current

- Optimizing GPU utilization in the ACB project by reducing idle time through workload balancing and scheduling strategies.
- Implementing multi-server configurations to enhance parallel processing efficiency and minimize resource underutilization.
- Exploring integration of agentic AI technologies into the Kojuchi platform to enable autonomous decision-making and workflow orchestration.

Machine Learning Researcher (Fellowship) MIT + Radius Collaboration, USA

USA

July 2025

- Designed and prototyped a **Radius Testnet trust and discovery layer** for agent marketplaces, integrating **smart contract-based reputation** and ranking mechanisms.
- Implemented **MCP server monetization workflows**, including payments, invoicing, and protected deployments with EVMAuth and TypeScript SDK adapters.
- Advanced research on **agentic AI ecosystems**, focusing on decentralized discovery, trust, and microtransaction-based interoperability for autonomous agents.

Machine Learning Engineer Akdene Technologies

USA

Mar 2025 – July 25

- Developed an AI-powered financial risk engine using TensorFlow, XGBoost, and PyTorch Geometric, deploying real-time fraud detection and credit scoring for 100K+ customers in <3 months
- Built an Agentic AI chatbot using GPT-4o, RAG, and AutoGen on Django, cutting support workload by 120+ hrs/month and slashing response time by 65% via real-time semantic caching and multi-agent automation.
- Deployed on serverless AWS with auto-scaling Kubernetes, achieving 99.9% uptime while reducing API costs by 30% through LoRA fine-tuning and Redis Vector Search.
- Enhanced fraud detection with PyTorch and model interpretability techniques, identifying 1,200+ high-risk transactions in the first weeks of deployment.
- Formulated a model monitoring pipeline using Conformal Predictions, ensuring 75% stability in model accuracy across financial datasets.

Software Engineer ServiceNow

India

Jun 2020 – Aug 2022

- Deployed a scalable backend microservice using Java, Spring Boot, and REST APIs within 6 months, improving data processing speed by 3x for enterprise clients.
- Implemented a secure authentication system using OAuth2, API Development, and system integration, reducing unauthorized access incidents by 60% in the first year.
- Accelerated cloud-based deployment on GCP, reducing infrastructure costs by 25% while maintaining 79% uptime.

Advanced Transformer Optimization for CHAI Model

Lead Researcher – US Meta Research Collaborator

Apr 2024

- Pioneered a cutting-edge hybrid sparse attention mechanism combined with precision fine-tuning across three domain-specific datasets, enhancing task accuracy by 15% while slashing inference time by 50%.
- Engineered ultra-efficient transformer pipelines with revolutionary memory optimization, reducing GPU memory consumption by over 4GB per training cycle and enabling seamless deployment on resource-constrained systems.
- Innovated AI-driven clustering techniques to streamline token embeddings, reducing pre-processing overhead by 40% and dramatically accelerating end-to-end model training.

RESEARCH PROJECTS

Differential Privacy of Multimodal Clinical Data - Prof. Praneeth Vepakomma

April 25 – July 25

- Developed a patient-specific differential privacy pipeline by generating synthetic clinical reports, applying DP-Prompt paraphrasing, and training privacy-preserving embeddings.
- Organized and optimized embeddings per patient to enable scalable privacy analysis across medical datasets.

EDUCATION

PhD in Computer Science

University of Missouri, Columbia

Jan 2026

Master of Science in Machine Learning and Data Science

University of California, San Diego, CA

June 2024

Bachelor's in Engineering

IIT Hyderabad, India

June 2020

TECHNICAL SKILLS

• Programming Languages	Python (Expert), Java, JavaScript, C++, R, Fortran, Prolog, Perl, Kotlin, Swift
• Machine Learning & AI	Machine Learning, Deep Learning, NLP, Federated Learning, Transformers, Multimodal AI, Model Interpretability, Conformal Predictions
• AI/LLM Tools	NLTK, Langchain, OpenAI, Google Gemini
• ML Libraries & Frameworks	Scikit-learn, TensorFlow, Keras, XGBoost, PyTorch, Django, Langchain
• Data Analysis & Visualization	Pandas, NumPy, Matplotlib, Seaborn, Dashboards, Pre-built Analytics Metrics
• Cloud & DevOps	Azure, GCP, Kubernetes, Git, CI/CD, Technical Documentation
• Databases & Data Tools	SQL, Elasticsearch, ETL
• API & Backend Development	REST APIs, API Development, Java APIs, Microservices, System Integration
• Development Practices	Agile, System Design, Testing, Model Monitoring, MLOps

PUBLICATIONS

- Building Communication Efficient Peer-to-Peer Federated LLMs with Blockchain, **AAAI, Stanford University** ([link](#))
- Explore-Then-Commit: The Optimal Strategy for Scientific Breakthrough Discovery, **AAAI 2025**
- Fortifying the Agentic Web: A Unified Zero-Trust Architecture Against Logic-layer Threats ([link](#))
- AILUMINATE: Introducing v1.0 of the AI Risk and Reliability Benchmark from MLCommons ([link](#))
- An Index-based Federated Registry Architecture for Secure, Capability-Aware Agent Discovery: Leveraging NANDA and ANS ([link](#))
- CPTQuant - A Novel Mixed Precision Quantization Techniques for Large Language Models ([link](#))
- Decoding Federated Learning: The FedNAM+ Conformal Revolution ([link](#))
- The Trust Fabric: Decentralized Interoperability and Economic Coordination for the Agentic Web ([Link](#))

AWARDS AND ACHIEVEMENTS

- Selected as Research Fellow for **MIT NANDA Radius Fellowship program 2025**
- **Skill development incentive program** award, **ServiceNow 2021**
- **Academic excellence award**, IIT Hyderabad 2018
- Selected as a peer reviewer for **NeurIPS 2025**
- Project accepted to the 2025 **NSF AI-SDM Workshop** on Human-AI Complementarity for Decision Making
- Co-Founded **AutoPatch+**, a next-gen AI code validation platform, and pioneered hallucination detection tech for generative coding, showcased at MIT AI Summit 2025 - [AutoPatch+](#)
- Developed Multimodal small LLM of size 125M parameters - [PixPrompt](#)
- Awarded a **Research Fellowship** with EleutherAI Summer of Open AI Research.
- Granted a **Full Scholarship and Research Fellowship** with Algoverse Open AI Research 2025.
- Chosen as a **Full Scholarship Recipient** for the PEARC25, 26 Student Program 2025.
- Accumulated **75+ citations across 11 research publications** on Google Scholar